Smaran Dhungana

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Prospective Ph.D. Applicant in Computer Science & Engineering — AI Safety, Distributed Systems & Verifiable Reasoning

Research Interests

- AI Agent Safety & Governance: Zero Trust identity architectures, Model Context Protocol (MCP) runtime verification, capability delegation, and mitigation of indirect prompt injection and cascading tool-use vulnerabilities.
- Resilient Distributed Systems: Multi-agent coordination protocols, computation/communication co-scheduling, state synchronization under partial observability, and scalable data infrastructure.
- Neuro-Symbolic & Verifiable AI: Grounding neural decision-making with explicit domain ontologies, formal knowledge graphs, and axiomatic constraint satisfaction models.

Education

Tribhuvan University, Institute of Engineering (IOE), Pulchowk Campus — Lalitpur, Nepal
Bachelor of Engineering (B.E.) in Computer Engineering — First Division (71.67%) — Dec 2018 – Dec 2022

– Institution Note: Pulchowk Campus is the flagship and most selective engineering college in Nepal (< 3% national entrance exam acceptance rate).

Shree Amar Singh Model Secondary School — Pokhara, Nepal
High School Diploma, Science Stream (Physics, Mathematics, Chemistry) — 2016 – 2018

Standardized Test Scores

- TOEFL iBT: 100 / 120 (CEFR Level C1) — Reading: 25/30, Listening: 27/30, Speaking: 20/30, Writing: 28/30.
- GRE General Test: 316 / 340 — Quantitative: 160/170, Verbal: 156/170, Analytical Writing: 3.5/6.0.

Research & Professional Experience

Astha.ai — Palo Alto, CA (Remote)
Senior AI Engineer — Aug 2025 – Present

– Lead research on security, runtime governance, and threat modeling for autonomous multi-agent tool execution and Model Context Protocol (MCP) ecosystems.
– Active core contributor to SAFE-MCP, an open-source framework under the Linux Foundation dedicated to auditing agent-tool orchestration vulnerabilities.
– Co-designed a Zero Trust Identity System for AI Agents, introducing fine-grained cryptographic capability tokens, continuous policy enforcement, and provenance tracking across tool boundaries.
– Formulated formal compliance mappings between SAFE-MCP safety controls and established global standards including NIST AI RMF, NIST SP 800-207, HIPAA, and GDPR.

Autolab Technologies Pvt. Ltd. — Kathmandu, Nepal
Lead Backend Engineer — Jul 2024 – Aug 2025

– Researched and implemented distributed data extraction and mining architectures for Japanese industrial retail leader Misumi Co.
– Architected high-performance spec-matching software for complex mechanical components, formulating multi-iteration geometric and semantic attribute search algorithms.
– Coordinated technical system architectures and interface contracts between bilingual engineering teams in Nepal and Japan.

Alpas Technology — Kathmandu, Nepal
Data Engineer — Sep 2023 – Jul 2024

– Designed and deployed an end-to-end multi-metric (100+ signals) search intelligence and analytics engine from the data ingestion layer to application APIs.
– Designed high-throughput columnar data warehouse schemas and recurring automated ETL DAGs handling time-windowed aggregations from 20+ to 100+ heterogeneous external sources.
– Applied first-principles axiomatic modeling to structure multi-level compound metrics into actionable business value indices.

Key Research & Engineering Projects

SAFE-MCP: Agent Safety & Vulnerability Auditing Framework | Linux Foundation, Python, MCP 2025 – Present
– Open-source safety framework establishing vulnerability taxonomies and testing suites for MCP servers, client tools, and autonomous agent loops.

Zero Trust Agent Identity Architecture | Astha.ai, Cryptographic Identity, Policy Enforcement 2025 – Present
– Engineered cryptographic verification and least-privilege scoping mechanisms to prevent unauthorized privilege escalation in multi-agent tool execution pipelines.

High-Throughput Mechanical Spec-Matching Engine | Autolab / Misumi Japan, Python, C++, Distributed Systems 2024 – 2025
– High-speed distributed spec-matching pipeline comparing millions of physical component specifications using geometric tolerances and semantic attribute matching.

Dataverse: Campus-Wide Data Science & ML Competition | LOCUS, Pulchowk Campus 2022
– Conceived, designed, and executed a major collegiate competitive data science event to foster competitive machine learning and practical data analysis among engineering students.

Undergraduate Capstone Engineering Project | Tribhuvan University, IOE Pulchowk Campus 2022
– Formulated and implemented full-cycle engineering capstone project in systems programming and applied machine learning evaluated by departmental faculty jury.

Relevant Academic Coursework (Pulchowk Campus)

- Systems & Networks: Distributed Systems, Computer Networks, Operating Systems, Computer Organization & Architecture, Data Communication.
- Computer Science Theory: Theory of Computation, Data Structures & Algorithms, Object-Oriented Analysis & Design, Software Engineering, Discrete Structures, Simulation & Modeling.
- AI & Data Systems: Artificial Intelligence, Big Data Technologies, Database Management Systems, Digital Signal Analysis & Processing.
- Mathematical Foundations: Applied Mathematics, Engineering Mathematics I, II & III, Numerical Methods, Probability & Statistics.

Technical & Theoretical Skills

- Programming Languages: Python (NumPy, SciPy, Pandas), C++, SQL, Bash, Go, LaTeX, HTML/CSS.
- AI & Agent Safety: Model Context Protocol (MCP), SAFE-MCP, Zero Trust Agent Architecture, Multi-Agent Systems, Tool Sandboxing, Threat Modeling, Prompt Injection Mitigation.
- Machine Learning & Data: PyTorch, TensorFlow, Scikit-learn, AutoML, Columnar Data Warehouses, Time Series & Sequence Models, ETL Pipeline Engineering.
- Systems & Infrastructure: Distributed Systems, Linux/Unix Internals, Docker, Git, RESTful APIs, gRPC, Microservices, CI/CD.
- Security Standards: NIST AI RMF, NIST SP 800-207 (Zero Trust), HIPAA Security Rule, GDPR Article 32 Controls.
- Theoretical Foundations: Axiomatic Modeling & Formal Reasoning, Theory of Computation, Ontologies & Knowledge Graphs, Discrete Mathematics, Probability & Statistics.

Academic Service, Leadership & Creative Work

- Software Coordinator, LOCUS (2022 Tenure): Elected executive member of the undergraduate engineering society at Pulchowk Campus; spearheaded campus software infrastructure, hackathons, and technical symposia.
- Playwright & Director: Wrote and directed an original full-length theatrical production for the annual campus cultural festival, recognized by campus consensus as the standout production of the event.
- Certifications: Analyze Datasets and Train ML Models using AutoML (Google Cloud), Structuring Machine Learning Projects (DeepLearning.AI), Sequences, Time Series and Prediction (DeepLearning.AI).